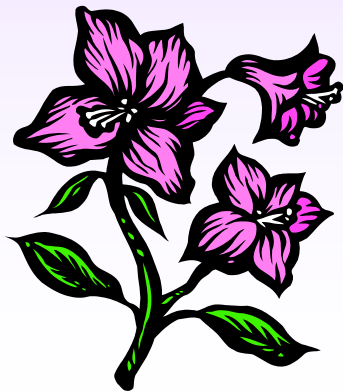




Buffer system

Acidosis & Alkalosis





Buffer system

Acidosis & Alkalosis



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By the end of this lecture, the student will be able to:

A1. Define acids, base, pH.

A2. Point out the regulatory mechanisms of acid base homeostasis

A3. Describe acid base disorders and their types, causes, compensatory mechanisms.

A4. Interpret arterial blood gases report.

Acid Base Balance

- **Human diet is almost neutral but the metabolic processes in the body produce large amounts of acids like carbonic , Citric, sulfuric, phosphoric....etc.**
- **These products are transported to the lung (CO_2) and kidney (sulfuric, phosphoric) via blood and ECF without changing arterial PH (7.35 - 7.45) and venous PH (7.32-7.38).**

This balance is owing to 3 important regulatory mechanisms:

**1- Buffering capacity of blood, React very rapidly
(less than a second)**

**2- Respiratory regulatory mechanisms. Reacts
rapidly (seconds to minutes)**

**3- Renal regulatory mechanisms. Reacts slowly
(minutes to hours)**

Effects of pH

The most general effect of pH changes are on **enzyme function, also hormones and ion distribution** are all affected by Hydrogen ion concentrations.

ACIDS

Acids can be defined as chemical substances that donate protons (H^+)

Strong acids readily give protons

Physiologically important acids include:

Carbonic acid (H_2CO_3)

Phosphoric acid (H_3PO_4)

Pyruvic acid ($\text{C}_3\text{H}_4\text{O}_3$)

Lactic acid ($\text{C}_3\text{H}_6\text{O}_3$)

Bases

**Bases can be defined as a substance that accept protons
(H⁺)**

Strong bases readily accept protons

Physiologically important bases include:

Bicarbonate (HCO₃⁻) and Bi-phosphate (HPO₄⁻²)



pH Meter for measurement of pH of solutions

How to use a Blood Gas Analyser



ABG apparatus for measurement of blood pH

Acid Base Balance

- Acid Base balance in the body is owing to three main systems:

1-Buffer system and their role in acid base regulation

A buffer is a mixture of weak acid and a salt of its conjugate base that resists changes in pH when a strong acid or base is added to solution.

Types of physiological buffers

1) Bicarbonate Buffer

2) Phosphate Buffer

3) Protein Buffer

A. Bicarbonate buffer system

Bicarbonate Buffer System HCO_3^- / H_2CO_3

Normal ratio of HCO_3^- to H_2CO_3 is 20:1

The chief buffer of blood

B. Phosphate buffer system

Phosphate buffer system HPO_4 / H_2PO_4

The chief buffer of urine.

C. Protein buffer system

Behaves as a buffer in both plasma and cells. **Albumin** is the most important in plasma. **Hemoglobin** is by far the most important protein buffer inside erythrocyte.

Abnormal Acid-Base Status

Acid base disturbances are traditionally classified into:

Metabolic acidosis

Metabolic alkalosis

Respiratory acidosis

Respiratory alkalosis

Normal values of Arterial Blood Gases (ABG)

PH 7.35 – 7.45

PaCO₂ (35 – 45) mmHg

HCO₃ (22 – 26) mEq /L

PaO₂ (80 – 100) mmHg

O₂ Saturation (95 % – 98)%

Acidosis

- **Acidosis** results from increasing hydrogen ion concentration of arterial blood plasma
- Generally acidosis occurs when *pH* of the blood decreases below 7.35
- Acidosis is either due to **Respiratory** or **Metabolic** causes

A. Respiratory acidosis

It results from any condition decreasing the rate of respiration = **hypoventilation**.

- In respiratory acidosis, the **CO₂** is **increased** while the **bicarbonate** is either **normal** or **increased**

Causes of respiratory acidosis:

1. Pulmonary problems e.g.:

Pneumothorax, emphysema, chronic bronchitis, asthma, severe pneumonia, and aspiration

2. Head injuries

3. Drugs (especially anaesthetics and sedatives)

4. Brain tumors

The response to acidosis as follows: By increasing renal reabsorption of NaHCO_3 to restore the ratio. increase

urinary excretion of H^+ , NH_4^+ and Cl^- to help reabsorption of NaHCO_3



B. Metabolic acidosis

- Metabolic acidosis results from the increased production of metabolic acids
- In metabolic acidosis, the CO_2 is normal while the bicarbonate is decreased

Causes of metabolic acidosis:

1. **Renal acidosis** is due to renal failure in which, there is decrease in the ability of kidneys to excrete acid with an accumulation of urea and creatinine.
2. **Lactic acidosis** may occur from: **Hypoxemia** and **Hypoperfusion**
3. **Starvation and diabetic ketoacidosis** is due to the accumulation of ketoacids (ketosis).
4. **Severe or chronic diarrhea.**

The kidneys response to acidosis as follows:

- Tubular cells **reabsorb more bicarbonate** from the tubular fluid to the blood
- Collecting duct cells **excrete more hydrogen** and **generate more bicarbonate**, and **increase formation of the NH_3** .

Alkalosis

- Alkalosis results from **reducing hydrogen ion concentration of arterial blood plasma.**
- Generally alkalosis occurs when pH of the blood exceeds 7.45
- Alkalosis is either due to **respiratory** or **metabolic** causes

A. Respiratory alkalosis

- It results from any condition increasing the rate of respiration = hyperventilation
- In respiratory alkalosis, the CO_2 is decreased
- There are two types of respiratory alkalosis: **acute** and **chronic**:
 1. **Acute respiratory alkalosis** occurs rapidly and the person may lose consciousness
 2. **Chronic respiratory alkalosis** is a more long-standing condition

Causes of respiratory alkalosis:

- 1. Excessive mechanical ventilation**
- 2. Psychiatric causes:** anxiety, hysteria and stress
- 3. CNS causes:** stroke, meningitis
- 4. Drug use:** aspirin, caffeine
- 5. High altitude areas,** where the low oxygen stimulates increased ventilation
- 6. Fever**
- 7. Pregnancy**

The lungs response to alkalosis as follows:

- ☐ By decreasing breathing, expelling less CO₂

B. Metabolic Alkalosis

Causes of metabolic alkalosis

- 1. Prolonged vomiting**, resulting in a loss of hydrochloric acid
- 2. Consumption of alkali**
- 3. Endocrine disorders such as Cushing's syndrome and associated with hypokalaemia.**

The kidneys response to alkalosis by:

- 1. Excreting more bicarbonate in urine**
- 2. Decreasing hydrogen ion excretion from the tubular epithelial cells**
- 3. Increasing glutamine synthesis**

Interpreting ABGs

- 1. Look at the pH: is the primary problem acidosis (low) or alkalosis (high)**
- 2. Check the CO₂ (respiratory indicator) :is it less than 35 (alkalosis) or more than 45 (acidosis)**
- 3. Check the HCO₃ (metabolic indicator) is it less than 22 (acidosis) or more than 26 (alkalosis)**
- 4. Which is primary disorder (Resp. or Metabolic)?**

Reference

- 1- Harpers illustrated Biochemistry
- 2- Lippincott illustrated reviews integrated systems

